

- (c) 497 (d) 1449
(e) None of these
137. $\frac{768 \times 768 \times 768 + 232 \times 232 \times 232}{786 \times 768 - 768 \times 232 + 232 \times 232} = ?$
(a) 1000 (b) 536
(c) 500 (d) 268
(e) None of these
138. $\frac{854 \times 854 \times 854 - 276 \times 276 \times 276}{854 \times 854 + 854 \times 276 + 276 \times 276} = ?$
(a) 1130 (b) 578
(c) 565 (d) 1156
(e) None of these
139. $\frac{753 \times 753 + 247 \times 247 - 753 \times 247}{753 \times 753 \times 753 + 247 \times 247 \times 247} = ?$
(a) $\frac{1}{1000}$ (b) $\frac{1}{506}$
(c) $\frac{253}{500}$ (d) None of these
140. $\frac{256 \times 256 - 144 \times 144}{112}$ is equal to (S.S.C., 2010)
(a) 420 (b) 400
(c) 360 (d) 320
141. If $a = 11$ and $b = 9$, then the value of $\left(\frac{a^2 + b^2 + ab}{a^3 - b^3} \right)$ is (S.S.C., 2010)
(a) $\frac{1}{2}$ (b) $\frac{1}{20}$
(c) 2 (d) 20
142. If $a + b + c = 0$, $(a + b)(b + c)(c + a)$ equals (M.C.A., 2005)
(a) $ab(a + b)$ (b) $(a + b + c)^2$
(c) $-abc$ (d) $a^2 + b^2 + c^2$
143. If $a = 7$, $b = 5$, $c = 3$, then the value of $a^2 + b^2 + c^2 - ab - bc - ca$ is
(a) -12 (b) 0
(c) 8 (d) 12
144. Both addition and multiplication of numbers are operations which are
(a) neither commutative nor associative
(b) associative but not commutative
(c) commutative but not associative
(d) commutative and associative
145. Which of the following digits will replace the H marks in the following equation?
 $9H + H8 + H6 = 230$
(a) 3 (b) 4
(c) 5 (d) 9
(e) None of these
146. Find the missing number in the following addition problem:
- $$\begin{array}{r} 8 \quad 3 \quad 5 \\ 4 \quad * \quad 8 \\ + \quad 9 \quad * \quad 4 \\ 2 \quad \underline{2 \quad * \quad 7} \end{array}$$
- (a) 0 (b) 4
(c) 6 (d) 9
147. What number should replace M in this multiplication problem?
- $$\begin{array}{r} 3 \quad M \quad 4 \\ \times \quad 4 \\ \hline 1 \quad 2 \quad 1 \quad 6 \end{array}$$
- (a) 0 (b) 2
(c) 4 (d) 8
148. If p and q represent digits, what is the maximum possible value of q in the statement (S.S.C., 2010)
 $5p9 + 327 + 2q8 = 1114$?
(a) 6 (b) 7
(c) 8 (d) 9
149. What would be the maximum value of Q in the following equation?
 $5P7 + 8Q9 + R32 = 1928$
(a) 6 (b) 8
(c) 9 (d) Data inadequate
(e) None of these
150. What should come in place of * mark in the following equation?
 $1*5\$4 \div 148 = 78$
(a) 1 (b) 4
(c) 6 (d) 8
(e) None of these
151. If $6*43 - 46@9 = 1904$, which of the following should come in place of *?
(a) 4 (b) 6
(c) 9 (d) Cannot be determined
(e) None of these
152. What should be the maximum value of Q in the following equation?
 $5P9 - 7Q2 + 9R6 = 823$
(a) 5 (b) 6
(c) 7 (d) 9
(e) None of these
153. In the following sum, '?' stands for which digit?
 $? + 1? + 2? + ?3 + ?1 = 21?$
(a) 4 (b) 6
(c) 8 (d) 9
(e) None of these
- Directions (Questions 154–155):** These questions are based on the following information:
 $CBA + CCA = ACD$, where A, B, C and D stand for distinct digits and $D = 0$.